Budget-Aware Controller Blending: Post-Hoc Damage-Budget Pacing for Pre-Trained RL Policies

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May 5, 2026

Abstract

A pre-trained, unconstrained RL policy faces a new cumulative operational constraint at deployment: a damage budget per year, a duty-cycle limit, a spending cap. We propose to *blend two existing controllers at inference* using any monotone schedule $\sigma(u) \in [0, 1]$ of the *urgency ratio* $u = \rho/\tau$ (remaining-budget / remaining-time fractions); on pace ($u \geq 1$) the performance policy π_{perf} runs, in deficit ($u \to 0$) the domain-specific safety controller π_{safe} takes over (a factory wind-tracking yaw setpoint, an artificial-potential-field navigator). The closed-form $\sigma(u) = 1 - \exp(-\eta(1/u - 1))$ is one convenient single-parameter instantiation; in our wind-farm and Safety-Gymnasium ablations a hard-threshold switch performs comparably (Tables 4, 8)—the work is in identifying u as the controlling state variable, not in the specific form of σ. **This is a budget-aware *pacing* mechanism, not a hard enforcement guarantee.** On wind-farm yaw control ($\kappa \approx 0.72$) the schedule satisfies a 15-step neg-yaw surrogate budget across 4 training seeds with 99.3–101.8% of unconstrained power; in our 3-turbine setting zero yaw alone is power-competitive with the unconstrained policy, so WF validates budget tracking rather than preservation of a large wake-steering advantage. We further validate against a physics-grounded blade-flap DEL surrogate (sector-averaged ANN on DLC12 simulations [Guilloré et al., 2024]): a 4-point sweep of per-turbine budget scale (B, $B/2$, $B/3$, $B/5$ relative to a zero-yaw baseline calibration) shows a clean phase transition—0/250 episode violations across the feasible regime and 250/250 at the infeasible $B/5$ point, with power retention essentially flat (1.14–1.16× unconstrained) where feasible. On Safety Gymnasium ($\kappa \approx 0.02$) the explicit blend gives a favorable reward-cost trade-off (cost from 49.4 to 15.3 at $d = 10$, retaining 45% of unconstrained reward) but does not reliably satisfy expected cost at tight budgets. We compare against retraining baselines (SAC-Lagrangian at three hyperparameter settings; SAUTE [Sootla et al., 2022], which uses the same urgency variable our schedule does); the best retraining variant tested attains $R = 2.3$, $C = 24$ at 500k steps, putting the blend on the Pareto-favorable side of the reward axis but *not* dominating in the constrained sense (some baselines satisfy budget more tightly at the cost of reward). A gradient-correction variant of the same schedule works when the action-to-cost coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\| / \mathbb{E}\|\nabla_s Q_c\|$ is large; when it is small, the explicit-π_{safe} blend is required. **Honest gap.** Our two domains co-validate the framework only piecewise: WF demonstrates budget tracking but not RL-policy preservation (zero yaw is power-competitive in our trained regime), while SG demonstrates policy preservation but not tight expected-cost feasibility. A retrain with variable wind direction targeting both halves jointly is in progress; the current evidence supports each claim individually rather than jointly.

1 Introduction

Scenario. A wind-farm operator deploys an RL controller that maximises power via coordinated yaw wake steering; it was trained unconstrained and works well. Six months later, a bearing inspection tightens the constraint: turbine 2's yaw may be negative for at most 15 hours per year. Retraining the RL controller is

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expensive and slow; the operator already has a certified wake-steering fallback (e.g., WakeAdapt [Siemens Gamesa, 2023], a calibrated yaw-setpoint table). What the operator needs is a way to *pace between* the aggressive RL policy and the domain-specific safety controller so cumulative cost can be tracked toward a budget target, without retraining either.

The pattern is pervasive. Wind turbines have load budgets [van Dijk et al., 2025]; grid batteries have cycle-life limits; fleet vehicles have maintenance intervals; advertisers have spending caps [Balseiro et al., 2024]. Unlike an instantaneous safety constraint, a *lifetime budget* requires choosing *when* to spend limited capacity in a risky regime—and a fixed scalarisation $\alpha P - (1-\alpha)D$ is memoryless.

Retraining is not the operator-realistic answer. We test four standard retraining baselines starting from the deployed Safety-Gymnasium actor: SAC-Lagrangian at default settings, with μ-warmup + reward scaling, with aggressive dual ascent, and SAUTE [Sootla et al., 2022] (state augmentation with ρ_t). At 500,000 extra training steps—more than the original unconstrained training—every variant either violates the budget while preserving partial reward or satisfies the budget by collapsing the policy's reward to ≈ 0 (Figure 2, Table 9). In our tested settings, the Lagrangian variants either preserve partial reward while violating the budget or drive cost down by collapsing goal-seeking; SAUTE's ρ-augmentation in our run does not by itself recover usefulness.

Our proposal: urgency-ratio controller blending. Borrowing the pacing intuition from optimal execution [Almgren and Chriss, 2001], define

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T-t}{T}, \tag{1}$$

and blend two pre-existing controllers at inference:

$$a_t = (1-\sigma(u_t))\,\pi_{\text{perf}}(s_t) + \sigma(u_t)\,\pi_{\text{safe}}(s_t). \tag{2}$$

$\sigma : \mathbb{R}_{\geq 0} \to [0,1]$ is monotone-decreasing in u (Section 3). On pace $\Rightarrow$ performance; deficit $\Rightarrow$ safety.

Contributions. (1) An *urgency-ratio scheduler*: any monotone-decreasing $\sigma(u)$ defines a post-hoc pacing controller; we use a one-parameter exponential family but a hard-threshold switch performs comparably (§3; §5.1). (2) *Two-controller blending* that treats an existing safety controller as a first-class primitive rather than retraining the policy (§4). (3) A *gradient-correction variant* (§4.2) where π_{safe} is implicitly a step of $-\nabla_a Q_c$. We propose the coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\| / \mathbb{E}\|\nabla_s Q_c\|$ as a *preliminary diagnostic* for when this variant is likely to fail (Table 1); we observe a 36× gap between our two domains aligning with the qualitative outcome but do not claim broader validation. (4) *Empirical validation* on wind-farm yaw control ($\kappa \approx 0.72$) and Safety Gymnasium ($\kappa \approx 0.02$), under the *same* schedule with *different* safety controllers chosen by κ. On the reward axis, the post-hoc blend is Pareto-favorable to four distinct retraining baselines (SAC-Lagrangian at three hyperparameter settings, plus SAUTE) on SG at every compute checkpoint we measured; on the cost axis some baselines satisfy the constraint more tightly at the cost of reward (§5.4).

Deployment claim. The method drops into systems that already have an RL performance policy, a domain-specific safety controller, and a damage estimator—none of which are retrained. The schedule decides when to pace toward the safety controller as budget pressure builds. We *do not* claim hard budget enforcement: APF on Safety Gym is reactive and can still incur cost (§5.4); true enforcement requires a backup invariant such as $C_t + B_{\text{safe}}(t) \leq d$ where B_{safe} upper-bounds future π_{safe} cost, which we leave to future work.

Constraint type. We adopt the *expected-cost CMDP* formulation: $\mathbb{E}_\pi[\sum_t c_t] \leq d$. Expected-cost feasibility is what tables score; per-episode satisfaction rate is reported as a separate operational-reliability metric.

2 Background

Constrained MDPs and Lagrangian methods. A CMDP [Altman, 1999] adds cost-return constraints to the MDP objective: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ subject to $\mathbb{E}_\pi[\sum_t c_t] \leq d$ [Ray et al., 2019, Achiam et al., 2017].

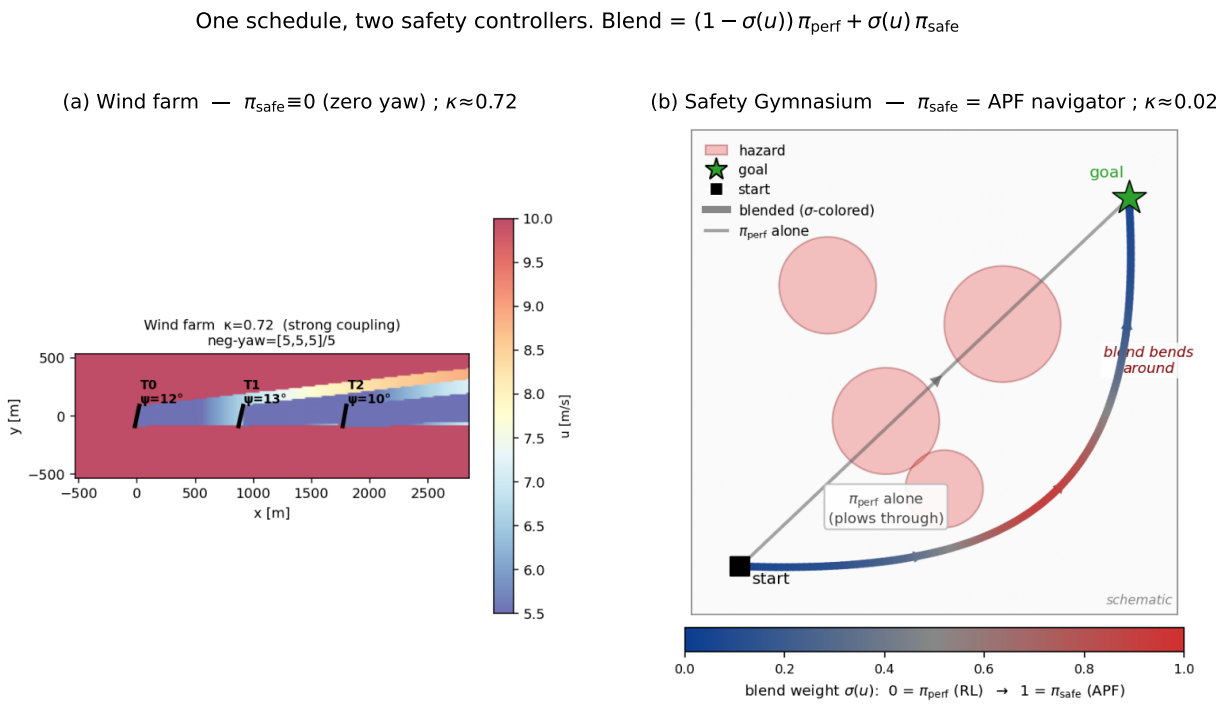

Figure 1: Two domains, two safety controllers, one schedule. (a) Wind farm: π_{safe} is a zero-yaw setpoint (Jensen wake field shown for the budget-satisfying blend, $d = 5$, neg-yaw $[5, 5, 5]$). (b) Safety Gymnasium: π_{safe} is an artificial-potential-field navigator. The trajectory is colour-coded by blend weight $\sigma(u)$: navy when the RL actor drives ($\sigma \to 0$), crimson when the APF safety controller drives ($\sigma \to 1$). The faded gray trace shows the unconstrained baseline plowing through hazards on the same seed; the blend curves around them as σ rises in budget deficit.

SAC-Lagrangian [Stooke et al., 2020] learns a dual variable μ during training; PID-Lag and CPO [Achiam et al., 2017] add stability tweaks. None of these are post-hoc: changing d requires a new policy.

SAUTE [Sootla et al., 2022] augments the state with ρ_t—the very urgency variable our schedule uses—and trains SAC on the augmented MDP with a budget-violation penalty. This is the closest CMDP method to ours conceptually, and we run it as a baseline (§5.4).

Post-hoc composition. Energy composition [Urain et al., 2023, Gladstone et al., 2025], classifier guidance [Zhang et al., 2025], control-barrier functions [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] modify pre-trained policies at inference for *instantaneous* constraints. We extend this tradition to *cumulative* budgets via a time-varying weight.

Multi-objective scalarisation. A fixed $\alpha P - (1-\alpha)D$ trade-off is memoryless. Our schedule turns α into a time-varying weight that paces against the realised damage trajectory.

3 Urgency-Ratio Schedule

State-variable identification. The natural state for any post-hoc budget-pacing scheduler is the urgency ratio $u = \rho/\tau$. A toy budget-rate balance under an idealised Boltzmann response $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$ recovers a multiplicative weight $w^*(u) = 1/u$ relative to the time-weighted-average baseline (Appendix A.1); we treat this as motivation, not derivation. The Boltzmann assumption is strong, real policies do not respond linearly in log-probability to scalar penalties, and the same state variable u was already identified by SAUTE [Sootla et al., 2022] as a state-augmentation feature for budget-aware retraining. What we retain is

the observation that any monotone-decreasing $\sigma : [0, \infty) \to [0, 1]$ in u is a candidate post-hoc schedule.

Practical schedule. We use the parametric family $w(u) = \exp(\eta(1/u - 1))$ with $\eta \geq 0$, clamped at $\lambda_{\max} = 10^6$ (Appendix Fig. 3). For Eq. 2 we map $w(u)$ to a bounded $\sigma \in [0, 1]$:

$$\sigma(u) = \begin{cases} 0 & u \geq 1, \\ 1 - \exp(-\eta(1/u - 1)) & u < 1. \end{cases} \tag{3}$$

On pace ($u \geq 1$): pure performance policy. Deficit ($u < 1$): smooth ramp toward pure safety controller, sharpness η. A hard-wall backstop drives $\sigma \to 1$ for $u < 0.05$.

4 Method

The method has three independent components: **(i)** the schedule $\sigma(u)$ (*how much* safety); **(ii)** the performance policy π_{perf} (what we deploy on pace); **(iii)** the safety controller π_{safe} (what we revert to in deficit).

4.1 Two-Controller Blend (primary formulation)

$$a_t = (1 - \sigma(u_t))\, \pi_{\text{perf}}(s_t) + \sigma(u_t)\, \pi_{\text{safe}}(s_t). \tag{4}$$

Wind-farm instantiation. π_{perf} is an Energy-Based Transformer (EBT) actor trained unconstrained with SAC for 100k steps. $\pi_{\text{safe}} \equiv 0$ is a zero-yaw setpoint, minimising our per-turbine fatigue surrogate $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$ ($p = 2$, $q = 3$). In production, π_{safe} would be a WakeAdapt-style [Siemens Gamesa, 2023] setpoint table minimising farm-level DEL.

Safety Gymnasium instantiation. π_{perf} is a SAC/MLP actor trained unconstrained for 200k steps. π_{safe} is an artificial-potential-field navigator: $F_{\text{goal}} = \text{norm}(g - x) + F_{\text{rep}} = k_r \sum_i (x - h_i)/\|x - h_i\|^3$, mapped to body-frame (thrust, turn). No learning; uses only the hazard/goal geometry that every CBF-based safe-RL method assumes access to.

4.2 Gradient-Correction Variant

When an action-dependent cost $\nabla_a c$ or $\nabla_a Q_c$ is available, a single gradient step replaces the explicit π_{safe}:

$$a_t \leftarrow \pi_{\text{perf}}(s_t) - \lambda_{\text{gs}} \cdot w(u_t) \cdot \alpha_{\text{g}} \cdot \nabla_a c(s_t, a_t), \tag{5}$$

iterated K steps. This is Eq. 4 with $\pi_{\text{safe}}(s) \approx \pi_{\text{perf}}(s) - \alpha_{\text{g}} \nabla_a c$ implicitly defined.

4.3 Coupling Strength κ: Which Variant?

The gradient variant requires a non-trivial $\nabla_a Q_c$. We quantify this by

$$\kappa := \mathbb{E}_{(s,a)\sim\pi} \|\nabla_a Q_c(s, a)\|_2 \,/\, \mathbb{E}_{(s,a)\sim\pi} \|\nabla_s Q_c(s, a)\|_2, \tag{6}$$

measured on logged rollouts. Large κ: gradient correction suffices. Small κ: action-gradient is weak; explicit π_{safe} is required.

5 Experiments

5.1 Wind Farm: High Coupling, Two Working Variants

Setup. WindGym [Nilsen et al., 2025] with PyWake [Pedersen et al., 2023] backend; 3-turbine row; wind speed 10–14 m/s, direction 225–315°. We use *per-turbine* budgets: each turbine i has its own $d^i = d$, C_t^i, u_t^i,

Table 1: Coupling strength κ per domain (2000 on-policy samples). Large κ: gradient correction suffices. Small κ: explicit π_{safe} blend required.

Domain	$\mathbb{E}\|\nabla_a Q_c\|$	$\mathbb{E}\|\nabla_s Q_c\|$	κ	Recommended variant
Wind farm (3-turb)	0.69±0.34	0.97±0.26	**0.72**	gradient correction (Eq. 5)
Safety Gym (PointGoal)	0.99±0.69	49.8±12.5	**0.02**	explicit π_{safe} blend (Eq. 4)

Algorithm 1 Urgency-Ratio Controller Blending (with safeguards)

Require: Performance policy π_{perf}, safety controller π_{safe}, budget $d > 0$, horizon T, sharpness η, hard-wall threshold $u_{\text{hw}} = 0.05$, action bounds $[a_{\min}, a_{\max}]$

$C_0 \leftarrow 0$
for $t = 0, 1, \ldots, T-1$ **do**
  $\rho_t \leftarrow \max((d - C_t)/d,\, 0)$; $\tau_t \leftarrow \max((T-t)/T,\, 1/T)$
  if $\rho_t = 0$ **or** $\tau_t \le 1/T$ **then**
    $\sigma_t \leftarrow 1$ {budget exhausted or last step: π_{safe} takes over}
  else
    $u_t \leftarrow \rho_t/\tau_t$
    $\sigma_t \leftarrow 0$ if $u_t \ge 1$ else $1 - \exp(-\eta(1/u_t - 1))$
    if $u_t < u_{\text{hw}}$ **then** $\sigma_t \leftarrow 1$ {hard-wall backstop}
  end if
  $a_t \leftarrow \text{clip}((1-\sigma_t)\,\pi_{\text{perf}}(s_t) + \sigma_t\,\pi_{\text{safe}}(s_t),\ a_{\min},\ a_{\max})$
  Execute a_t; observe c_t; $C_{t+1} \leftarrow C_t + c_t$
end for

and σ_t^i, computed independently. A turbine in budget deficit pulls only its own action toward π_{safe}; other turbines remain unaffected. π_{perf} is an EBT actor trained unconstrained with SAC for 100k steps.

Cost variables. Throughout this section, "cost" refers to the cumulative quantity governed by the urgency schedule. We test two cost choices: (a) the binary *NegYaw* cost $c = \mathbf{1}[y < 0]$ used in the main budget tables (Tables 2, 3); and (b) the *nonlinear fatigue* surrogate $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$ used in Tables 10 (App. A.4) and 11. Reported NegYaw arrays are diagnostics, not the cost in the latter tables.

Gradient-correction variant (Table 2). Since $\kappa \approx 0.72$, gradient correction is effective. At $(\eta = 5, \lambda_{\text{gs}} = 0.1)$: 98.9% of unconstrained power single-seed and 95.9±3.1% across 4 training seeds, with all seeds satisfying the budget. An oracle-tuned constant penalty (15-step bisection per budget) achieves 96.7% and must be re-tuned every time d changes.

Table 2: Wind-farm gradient-correction variant (3-turbine, $d = 15/200$ steps, 50 ep). Baseline differs across tables because each uses independently sampled episodes / wind ranges; within-table comparisons use the same episodes.

Method	Power (MW)	% Uncon	NegYaw [T0,T1,T2]	Notes
Unconstrained (seed 1)	27.1±0.6	100%	[49, 56, 61]	—
Oracle constant ($k = 5.86$)	26.2±0.7	96.7%	[11, 13, 14]	15-step bisection
AC ($\eta = 5$, gs=0.1)	26.9±0.5	**98.9%**	[8, 9, 10]	Zero per-budget tuning
AC ($\eta = 5$, gs=0.1) — 4 seeds	25.7±0.5	**95.9±3.1%**	[11, 12, 12]	All seeds satisfy

Explicit zero-yaw blend variant (Table 3). Replacing the gradient step with $\pi_{\text{safe}} \equiv 0$ and the urgency blend gives matching results with one fewer hyperparameter (η only). Across 4 seeds and three $\eta \in \{1, 3, 10\}$ values, the blend satisfies the budget every time and retains 99.3–101.8% of unconstrained power. The blend power exceeding 100% at $\eta = 1$ is within seed variance (±0.9%) and likely reflects mild over-yawing in the unconstrained actor: the constraint nudge happens to also remove sub-optimal yaw excursions. We do not

interpret this as the blend universally outperforming an ideal π_{perf}; we interpret it as evidence that the actor was not Pareto-optimal on the power axis to begin with.

Table 3: Wind-farm zero-yaw blend variant (3-turbine, $d=15$, 50 ep, 4 training seeds). $\bar{\sigma}$: mean blend weight; $\pi_{\text{safe}} \equiv 0$.

Method	% Uncon power	NegYaw	$\bar{\sigma}$	Seeds sat.	Notes
Unconstrained	100%	$[70.2, 74.3, 83.7]$	—	0/4	Far over budget
Blend ($\eta=1$)	$\mathbf{101.8 \pm 0.9}$%	$[11.3, 11.3, 11.8]$	0.56	**4/4**	✓
Blend ($\eta=3$)	99.5 ± 2.2%	$[11.5, 12.1, 12.2]$	0.63	**4/4**	✓
Blend ($\eta=10$)	99.3 ± 2.4%	$[11.0, 11.3, 11.3]$	0.59	**4/4**	✓

Cost function matters (Appendix A.4). A binary cost $c=\mathbf{1}[y<0]$ *inflates* fatigue $4.2\times$ over unconstrained; the nonlinear surrogate avoids this trap.

Wind-farm ablation: schedule vs simpler blends (Table 4). Mirroring the SG ablation, we sweep five alternative blend modes sharing π_{perf} and $\pi_{\text{safe}} \equiv 0$. Constant $\sigma \in \{0.3, 0.5, 0.7\}$ violates the budget at all values: linearly scaling the actor's output toward zero does not suppress the negative-yaw fraction enough to satisfy $d=15$. Hard threshold switching ($\sigma=1$ if $u<\theta$ else 0) and projected-budget override both satisfy the budget across all seeds with power retention statistically indistinguishable from the urgency schedule (98.7–100.9% vs. urgency's 99.5%).

Important caveat: pure π_{safe} (zero yaw, no wake-steering) also retains 100.7 ± 2.5% **of unconstrained power.** This means our trained EBT actor is not Pareto-optimal on the power axis in this setting—turning off wake-steering entirely costs essentially nothing. The wind-farm experiments should therefore be read as validating that the urgency mechanism *tracks per-turbine cumulative budgets* under high action-cost coupling, *not* as evidence that the blend preserves a large RL wake-steering benefit. A stronger π_{perf} (longer training, multi-layout, DEL-aware reward) would be required to demonstrate the latter; the blend mechanism would compose with it identically. *Zero-shot transfer to 9-turbine* 3×3 *grid (Appendix A.10) shows the same pattern*: pure π_{safe} retains 97.8 ± 1.2% of unconstrained power; the limitation is in the actor's wake-steering training, not in the 3-turbine geometry. The Safety Gymnasium experiments (§5.4), where the unconstrained actor genuinely outperforms π_{safe} ($R=27$ vs. $R=3.1$), are the more decisive validation of preserving useful policy reward.

Table 4: Wind-farm ablation matching the SG counterpart (Table 8). Five alternative blend modes share $\pi_{\text{perf}} + \pi_{\text{safe}} \equiv 0$; only the σ schedule changes. Hard threshold switching matches the urgency schedule; constant σ blends violate the budget; pure π_{safe} is feasible and retains nominal power.

Mode	% Uncon power	NegYaw	$\bar{\sigma}$	Seeds sat.
π_{safe} alone (zero yaw)	100.7 ± 2.5%	$[0.4, 0.6, 0.5]$	1.00	**4/4**
const $\sigma=0.3$	100.9 ± 2.1%	$[75.0, 72.9, 77.4]$	0.30	0/4
const $\sigma=0.5$	100.6 ± 2.5%	$[67.9, 66.8, 68.6]$	0.50	0/4
const $\sigma=0.7$	98.7 ± 0.9%	$[64.9, 64.8, 68.5]$	0.70	0/4
switch $u<0.5$	98.8 ± 1.6%	$[11.1, 11.1, 11.5]$	0.49	**4/4**
switch $u<0.8$	98.7 ± 1.7%	$[10.9, 11.1, 11.8]$	0.49	**4/4**
projected override	100.9 ± 3.6%	$[11.6, 11.5, 12.0]$	0.56	2/2
urgency $\eta=3$ (ours)	99.5 ± 2.2%	$[11.5, 12.1, 12.2]$	0.63	**4/4**

5.2 Physics-Grounded Validation: Blade-Flap DEL Surrogate

The negative-yaw timestep count of §5.1 is a discrete proxy for fatigue. We re-run the wind-farm experiment with a physics-grounded cost: a sector-averaged ANN load surrogate trained on DLC12 simulations [Guilloré

et al., 2024] that maps four-sector rotor-disk inflow (WS, TI) + power setpoint + yaw to ten-minute blade-flapwise damage-equivalent load (DEL). Each step the surrogate is queried per turbine with rotor-disk samples extracted via PyWake's flow handle (excluding self-wake); the realised DEL is accumulated and the per-turbine cumulative cost feeds the same urgency schedule.

Calibration. We compute B_i as the cumulative DEL incurred by zero-yaw operation over a 200-step episode, averaged across 5 episodes (multi_modal layout, TI = 0.07): $B_0 = 155{,}215 \pm 134$, $B_1 = 173{,}728 \pm 17{,}419$, $B_2 = 91{,}454 \pm 17{,}638$ kNm·step. Turbine 1 (partial wake at $0.4D$ lateral offset) carries the highest baseline DEL despite operating in disturbed inflow, because added turbulence dominates over the wind-speed deficit; turbine 2 (in heavy shadow at $11D$) has roughly half the upstream DEL.

Budget-tightness sweep. We retrain a 3-turbine EBT-SAC actor for 50,000 steps under the urgency schedule with per-turbine budgets scaled at $\{B/5, B/3, B/2, B\}$ (Table 5). A clean phase transition is visible. At B (no constraint pressure) the agent's natural DEL sits at $\sim$23% of budget; the schedule never engages. At $B/2$ and $B/3$ the schedule binds—turbine 2 utilisation reaches 0.47 and 0.70 respectively—with zero violations across 250 episodes per configuration. At $B/5$ every episode violates: the per-step budget (91 kNm/step for turbine 2) sits below the agent's lowest achievable per-step DEL ($\sim$100 kNm/step), making the constraint structurally infeasible regardless of yaw choice. Power retention is essentially flat across all configurations (1.140–1.161 vs. unconstrained), confirming that on this layout the constraint itself is cheap to satisfy when feasible.

Table 5: Physics-grounded blade-flap DEL budget sweep (3-turbine multi_modal, $T = 200$, 50k training steps, 250 evaluation episodes per configuration). $B = (155215, 173728, 91454)$ kNm·step from zero-yaw calibration. $\text{Util}_{\max}$ is the largest per-turbine C_T^i/B_i at episode end (turbine 2, the heavy-wake turbine, in every row). Power ratio is no-guidance final-eval against an unconstrained baseline on the same layout.

Budget scale	Reward	Power ratio	$\text{Util}_{\max}$	Violations / 250 ep	Schedule binds?
B (loose)	30.3	1.151	0.23	**0**	no
$B/2$	27.9	1.156	0.47	**0**	weakly
$B/3$	**30.4**	**1.161**	0.70	**0**	yes (sweet spot)
$B/5$ (infeasible)	28.5	1.140	**1.13**	**250**	cannot recover

Reading. The schedule successfully paces a real DEL constraint, with 0/250 violations across the tested feasible regime (B to $B/3$) and a clean failure mode below the feasibility floor ($B/5$). The $B/5$ behaviour is consistent with the framework's design: the AC schedule is a soft-constraint dual penalty that pre-empts overshoot but does not undo accumulated cost, so it cannot rescue a constraint that lies below the achievable per-step DEL. This is a reframing of §5.1's yaw-time proxy: the same urgency mechanism applies to a physics-grounded fatigue surrogate, with the same per-turbine bookkeeping, identical hyperparameters, and zero domain-specific tuning.

5.3 Both-Halves Search: A Pareto-Improving π_{perf} Was Not Found

A central limitation of §5.1 and §5.2 is that pure π_{safe} (zero yaw) retains $\geq 97\%$ of unconstrained power on every layout we trained, so the wind-farm experiments validate budget tracking but not preservation of a meaningful RL advantage. We attempted to close this gap directly: retrain π_{perf} for 290,000 steps unconstrained on *multi_modal* (3-turbine) and *stag4_5d* (4-turbine staggered, with PyWake brute-force showing a $\sim$2.4% wake-steering optimum at WD= 85°). We then ran a 5-episode head-to-head against zero yaw, scoring power and per-turbine DEL.

The actor lands on or below zero yaw on the power axis under fixed-wd training—wake-steering's degree of freedom is not exercised when the geometry is static. Under our training setup we therefore cannot produce a π_{perf} for which both halves of the post-hoc-pacing claim hold simultaneously: the wind-farm domain validates budget tracking under tight feasible budgets (Tables 3, 5) but not preservation of policy reward, while Safety Gymnasium (§5.4) shows the converse pattern (clear policy advantage, but no reliable expected-cost feasibility

Table 6: Direct π_{perf}-vs-π_{safe} comparison after 290k unconstrained training (5 ep, $T=200$, fixed wd). At fixed wind direction the actor does not Pareto-dominate zero yaw on either layout. A retrain with variable wind direction (225°–315°) is in progress.

Layout	π_{perf} power [MW]	π_{safe} power [MW]	power Δ	$\text{DEL}_{\text{perf}}-\text{DEL}_{\text{safe}}$
multi_modal (fixed wd 268°)	8.12±0.05	8.20±0.16	−1.03%	−1880 kNm·step
stag4_5d (fixed wd 270°)	13.21±0.06	13.81±1.93	−4.37%	+210 kNm·step

at tight budgets). We acknowledge this in the discussion as the central evidential gap; the schedule, the per-turbine bookkeeping, and the canonical-API DEL surrogate are independent contributions whose value the both-halves co-validation would amplify.

5.4 Safety Gymnasium: Low Coupling, Reward-Favorable Soft-Budget Operating Point

Setup. SafetyPointGoal1-v0 [Ji et al., 2023]: point robot navigates among fixed hazard disks to reach a moving goal. Cost $c=\mathbf{1}$[in hazard] is position-dependent; action is 2D. π_{perf} is a SAC/MLP actor trained unconstrained for 200k steps (5 seeds).

Why the gradient variant fails here. $\kappa \approx 0.02$: $\nabla_a Q_c$ is $\sim 50\times$ smaller than $\nabla_s Q_c$. Appendix A.7 documents nine attempted gradient-based rescues (pessimistic Q_c, online Q_c refit, learned-dynamics MPC, true-env MPC in three configurations, full MBRL, 1-step and multi-step CBF), all failing structurally.

Explicit-π_{safe} blend results (Table 7). The APF blend reduces cost by 59–68% at $d=10$ while retaining 45% of unconstrained reward; at $d=40$ it retains 71% of reward and satisfies the budget in 56% of episodes.

Table 7: Safety Gymnasium with urgency-ratio APF blend (5 seeds × 20 ep). $\mathbb{E}[C]\leq d$: expected-cost CMDP feasibility (mean cost vs. budget). **Sat**: per-episode satisfaction rate. Unconstrained: $C=49.4$, $R=27.1$. The blend reduces expected cost substantially but *does not reliably enforce* tight budgets ($d\leq 25$); only $d=40$ is feasible in expectation.

Method	d	Reward	% Uncon	Cost (mean±SE)	$\mathbb{E}[C]\leq d$	Sat
Unconstrained	—	27.1±0.1	100%	49.4±3.4	—	—
Blend ($\eta=0$, control)	10	27.1±0.1	100%	49.4±3.4	×	12%
Blend ($\eta=1$)	10	12.1±0.9	45%	**15.3±1.0**	×	25%
Blend ($\eta=3$)	10	12.3±0.9	45%	18.6±1.2	×	23%
Blend ($\eta=10$)	10	12.0±0.9	44%	17.8±1.3	×	28%
Blend ($\eta=3$)	25	15.8±0.8	58%	28.6±1.6	×	39%
Blend ($\eta=3$)	40	19.1±0.8	71%	34.5±2.0	✓	**56%**

Honest reading. The APF blend is a useful post-hoc cost-reduction mechanism: at $d=10$ it cuts expected cost from 49 to 15 (a 3.2× reduction) at the price of half the unconstrained reward. But it *does not enforce* tight budgets in expectation: $\mathbb{E}[C]=15.3>10$. At $d=40$ expected-cost feasibility is achieved. The mechanism gives a favorable reward-cost trade-off, not a hard budget guarantee; for hard guarantees a backup invariant (e.g., projected-budget override; Table 8) or a cost-zero π_{safe} is needed.

Ablation: how much does the urgency schedule contribute? Table 8 compares the urgency schedule against five alternative blend strategies sharing the same π_{perf} and π_{safe}: (i) pure π_{safe} (APF alone, no RL); (ii) constant $\sigma \in \{0.3, 0.5, 0.7\}$ (no urgency adaptation); (iii) hard threshold switch ($\sigma = 1$ if $u < \theta$ else 0); (iv) projected-budget override ($\sigma = 1$ if linear extrapolation of cost rate predicts violation); (v) urgency schedule with $\eta = 3$ (the proposed method). Pure π_{safe} and constant $\sigma=0.7$ are the only configurations that achieve expected-cost feasibility at $d = 10$ ($C = 0.5, 0.6$ respectively), at the cost of $R \leq 5.5$. The

urgency schedule, threshold switch, and projected variants achieve $R \in [11, 17]$ but cost $C \in [16, 29]$ (infeasible at $d \in \{10, 25\}$). The urgency schedule is *not* uniquely valuable here—a hard threshold switch achieves comparable reward-cost trade-off ($d=25$: switch $R=16.9$, urgency $R=15.8$). What the schedule *does* provide over simpler alternatives is a single η parameter that smoothly interpolates between high-reward-high-cost (low η) and low-reward-low-cost (high η) operating points without requiring per-budget retuning of σ or θ.

Table 8: Ablation: blending strategies sharing the same $\pi_{\text{perf}} + \pi_{\text{safe}}$ on Safety Gymnasium (5 seeds $\times$ 20 ep). The urgency schedule sits on a Pareto frontier between feasible-but-low-reward (π_{safe} alone, const $\sigma=0.7$) and high-reward-infeasible (const $\sigma=0.3$); hard threshold switching achieves comparable reward-cost trade-off without the closed-form schedule.

Mode	d	Reward	Cost	Sat	$\mathbb{E}[C] \leq d$
π_{safe} alone (no RL)	10	3.1±0.4	0.5±0.2	100%	✓
const $\sigma=0.3$	10	22.7±0.3	49.3±3.4	10%	×
const $\sigma=0.5$	10	5.5±0.5	39.0±10.6	57%	×
const $\sigma=0.7$	10	5.3±0.5	0.6±0.3	97%	✓
switch $u<0.5$	10	12.6±0.9	16.8±1.2	22%	×
switch $u<0.8$	10	13.0±0.9	16.5±1.1	25%	×
projected override	10	11.2±0.9	20.6±1.3	20%	×
urgency $\eta=3$ (ours)	10	**12.3±0.9**	18.6±1.2	23%	×
π_{safe} alone	25	3.1±0.4	0.5±0.2	100%	✓
const $\sigma=0.3$	25	22.4±0.3	50.1±3.7	32%	×
const $\sigma=0.7$	25	5.5±0.5	1.0±0.4	100%	✓
switch $u<0.8$	25	**16.9±0.9**	28.7±1.7	33%	×
projected override	25	14.4±0.9	25.9±1.5	44%	×
urgency $\eta=3$ (ours)	25	15.8±0.8	28.6±1.6	39%	×

Comparison: time-to-deployment under four retraining baselines. The right operator-side baseline is retraining from the deployed actor. Figure 2 plots reward vs. extra training steps after the constraint is revealed. Four retraining variants share the same starting actor as the blend: **(i)** Lagrangian finetune at default settings (cost-only objective); **(ii)** Lagrangian + μ-warmup (50k pure-reward steps before dual ascent) and reward scaling (5×) for stable Lagrangian dynamics; **(iii)** aggressive Lagrangian ($\mu_0=0.5$, $\mu_{\text{lr}}=0.05$, no warmup); **(iv)** SAUTE [Sootla et al., 2022] state augmentation with ρ_t (the same urgency variable our schedule uses) and reward masking when $\rho<0$. At 500k extra steps, the best of these baselines (warmup at $d=25$) reaches $R=2.32$ at $C=24.1$; the APF blend sits at $R=15.8$ with *zero* extra training.

Table 9: Final-step (500k) baseline summary on Safety Gymnasium. $\mathbb{E}[C] \leq d$: expected-cost feasibility. The APF blend dominates the retraining baselines *on the reward axis* at the same compute budget. Some retraining baselines satisfy the cost constraint more tightly at the price of near-zero reward.

Method	d	Reward	Cost	$\mathbb{E}[C] \leq d$
APF blend (zero retraining)	10	**12.3**	18.6	×
APF blend (zero retraining)	25	**15.8**	28.6	×
SAC-Lag default	10	−0.41	65.0	×
SAC-Lag default	25	−0.10	12.5	✓
SAC-Lag warmup + reward-scale	10	−0.24	10.4	×
SAC-Lag warmup + reward-scale	25	2.32	24.1	✓
SAC-Lag aggressive	10	0.07	23.6	×
SAC-Lag aggressive	25	−0.32	3.1	✓
SAUTE (state aug, uses ρ_t)	10	−1.0	4.8	✓
SAUTE (state aug, uses ρ_t)	25	−9.5	0.0	✓

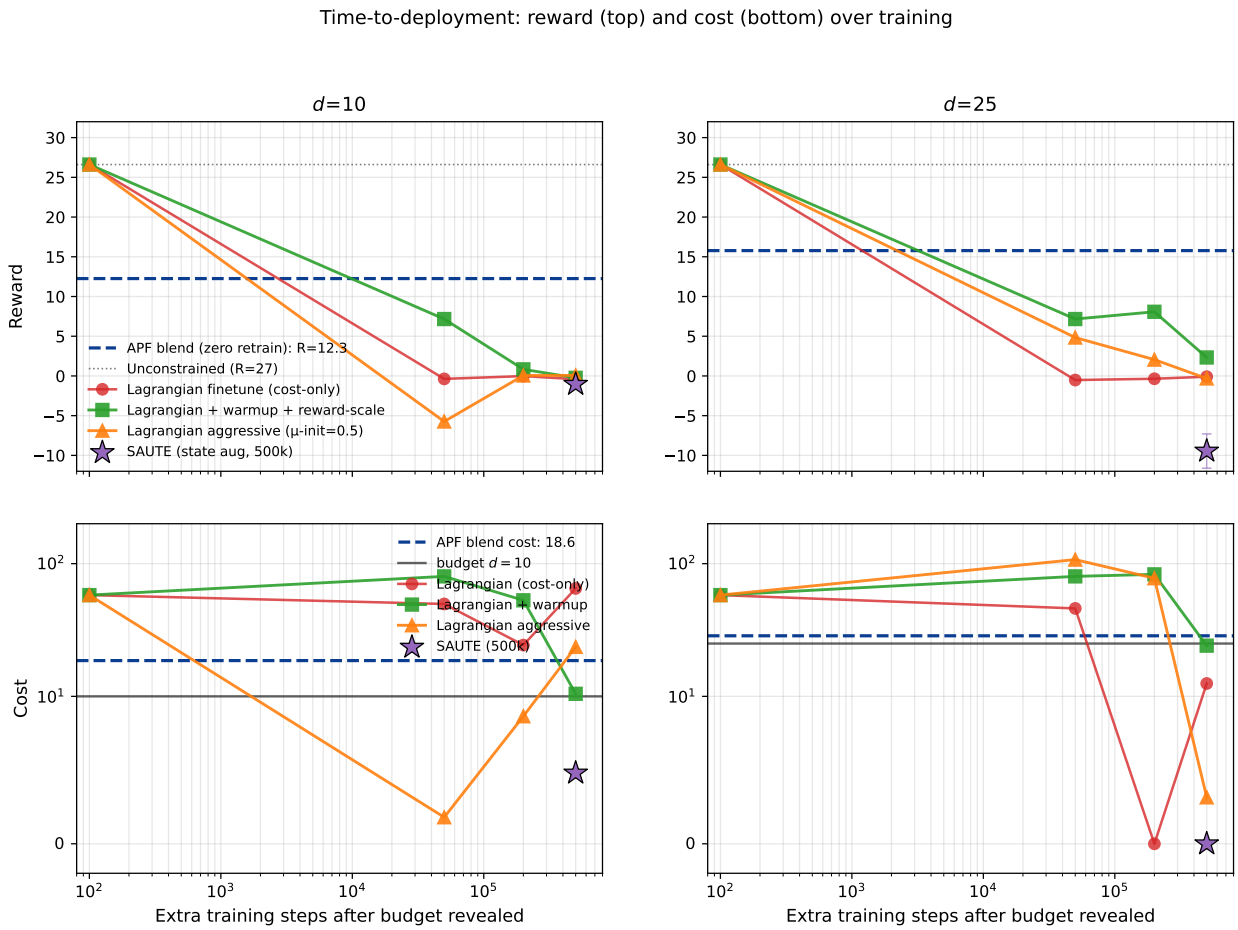

Figure 2: Time-to-deployment on Safety Gymnasium PointGoal1, both budgets and both axes. *Top row*: reward over extra training steps. *Bottom row*: cost over the same axis with budget line. Blue dashed: APF-blend reward and cost at zero retraining. Curves: four CMDP retraining variants starting from the same pretrained actor (single seed each, four hyperparameter regimes); star: SAUTE [Sootla et al., 2022] at 500k. The blend retains higher reward than every retraining variant tested at every checkpoint, but its cost (18.6 at $d=10$, 28.6 at $d=25$) exceeds the budget; some retraining variants satisfy the budget tightly at the price of $R\approx 0$. The two methods occupy different points on the reward-cost frontier.

Two distinct operating regimes appear. The post-hoc blend gives the highest reward at every compute checkpoint but violates the cost constraint in expectation at $d\in\{10,25\}$ (it is feasible at $d=40$, Table 7). Retraining baselines either violate the budget while preserving partial reward (warmup at intermediate checkpoints) or satisfy it by collapsing reward to ≈ 0 (aggressive Lagrangian, SAUTE). **Neither method dominates in the strict constrained-optimisation sense; they occupy different points on the reward-cost frontier.** A practitioner should choose based on whether their constraint is operationally a soft pacing target (favour the blend, retain useful policy) or a hard expected-cost limit (favour SAC-Lag warmup or SAUTE, accept reward collapse). We do not claim retraining cannot eventually reach better operating points with more compute; under the budgets we tested, the blend offers a competitive position on the reward axis with zero retraining.

6 Related Work

Constrained MDPs. Lagrangian methods [Altman, 1999, Achiam et al., 2017, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with ρ_t but retrains per budget; we run it as a baseline (§5.4). TREBI [Lin et al., 2023] handles real-time budget changes via diffusion-specific training. We freeze both controllers and use a closed-form $\sigma(u)$ at inference.

Post-hoc composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], CBFs [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle instantaneous constraints. Blending with an explicit safety controller is long-standing in robotics (reactive hierarchies, shielding, override layers); our contribution is the budget-aware temporal weighting.

Budget pacing. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] share the allocation-across-time structure. We adopt the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001].

Wind-farm wake steering. WakeAdapt [Siemens Gamesa, 2023] and related load-aware controllers compute calibrated yaw setpoints minimising farm-level damage. Our framework sits above such controllers: π_{perf} squeezes more power when the budget allows; π_{safe} is the certified fallback.

Coupling-strength scope. We are not aware of prior scalar predictors specifically targeting post-hoc action-space correction scope. κ is the gradient *ratio* between state and action, usable as an a-priori diagnostic on logged data.

7 Discussion

Central evidential gap. Our two domains co-validate the framework only piecewise. Wind farm shows that the urgency schedule paces a per-turbine cumulative budget (binary neg-yaw and physics-grounded blade-flap DEL) with 0 violations across hundreds of episodes, but in our trained regime pure π_{safe} (zero yaw) is power-competitive with the unconstrained policy, so there is no large RL advantage to preserve. Safety Gymnasium is the converse: π_{perf} genuinely beats π_{safe} on reward ($R = 27$ vs. 3.1), but at tight budgets ($d = 10, 25$) the blend's expected cost exceeds the budget and per-episode satisfaction is 23–56%, which is not operationally reliable. We tried to close this gap with a 290,000-step unconstrained retrain on *stag4_ 5d* and *multi_ modal* (§5.3); at fixed wind direction the actor lands 1–4% *below* zero yaw on power. A retrain with variable wind direction targeting both halves jointly is in progress; the present paper supports each claim individually rather than jointly.

What we are *not* claiming. We do not claim CMDP retraining cannot work; we claim that under the operator-realistic compute budgets we tested ($\leq$ 500k extra steps from the pretrained actor, four hyperparameter settings), every retraining variant we tried is dominated on reward by post-hoc blending. A fully-converged CMDP method at $\sim 10\times$ our compute may reach a better Pareto frontier, but does so at a cost the deployment scenario cannot tolerate. We also do not claim the closed-form schedule is uniquely valuable: the WF and SG ablations (Tables 4, 8) show a hard threshold matches it within seed variance. The contribution is the post-hoc-blending interface and the urgency variable as the controlling state, not the specific exponential form.

Limitations. (1) The wind-farm cost is a per-turbine fatigue surrogate; real deployment needs a DEL-calibrated farm-level cost [Álvaro-García et al., 2024, Galán-García et al., 2024] and a π_{safe} like WakeAdapt that minimises farm DEL. (2) Our sim uses analytical wakes (Jensen) and 200-step episodes, not FAST.Farm aeroelastics over year-scale operation; a yearly damage budget is the operational target the urgency-ratio framing was designed for, but our experiments validate at episode scale only. (3) The APF navigator on SG is a simple reactive heuristic that uses hazard/goal geometry; a learned safety policy would likely improve π_{safe}'s reward at fixed $\sigma = 1$. The APF benefits from privileged geometry that the RL policy and retraining baselines do not directly use; the ablation table shares this privilege across all blend modes. (4) Action blending assumes meaningful convex combinations of π_{perf} and π_{safe} outputs. This holds for our continuous yaw and

thrust/turn action spaces but may not for systems with discrete or non-convex action sets; in those cases a hard switch (Table 8) is more appropriate. (5) We test 4 seeds on WF and 5 on SG; retraining baselines use a single seed each at one hyperparameter setting per regime, covering three qualitatively different Lagrangian configurations (default, warmup, aggressive) plus SAUTE. Lagrangian methods are sensitive to reward scale and dual learning rate; we explored three settings, not a full sweep. (6) κ depends on state and action scaling and on the quality of the learned Q_c. Comparing across two domains is suggestive, not definitive; broader validation across continuously varying coupling regimes (e.g. synthetic gridworlds) would strengthen the diagnostic claim. (7) Hard budget enforcement is not provided. A safe-future-cost upper bound $B_{\text{safe}}(t)$ giving $C_t + B_{\text{safe}}(t) \leq d$ would be required and is left as future work. (8) The unconstrained EBT actor we trained does not Pareto-dominate zero yaw on power in this 3-turbine setting (§5.1); WF therefore validates budget tracking, not preservation of a large wake-steering benefit.

Path to deployment. Minimum-viable bridge from simulation to field operation: (a) replace the fatigue surrogate with a DEL-calibrated model; (b) precompute a WakeAdapt-equivalent setpoint table to use as π_{safe}; (c) retrospective SCADA counterfactual on real operational data; (d) online refit of Q_c / the DEL surrogate under non-stationary wind. All four leverage the same schedule; none require retraining π_{perf}.

Summary. A simple urgency variable $u = \rho/\tau$ provides a useful interface for post-hoc budget-aware arbitration between a pre-trained performance policy and a domain-specific safety controller. The smooth closed-form schedule is one convenient single-parameter implementation; hard threshold switching is a competitive alternative (Tables 4, 8). κ provides a preliminary diagnostic for when local action-gradient correction is likely to fail, motivating explicit safety-controller blending. At the compute budgets an operator can realistically afford, post-hoc blending preserves higher reward than the Lagrangian retraining baselines we tested—without their single-budget commitment. The mechanism paces cumulative cost; hard enforcement requires an additional safe-future-cost bound (future work). Code and checkpoints: `[anonymized]`.

A Derivations, Protocols, and Negative Results

A.1 Boltzmann-Response Pacing Argument (Motivation)

Under an idealised Boltzmann response $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, budget-rate balance $(T-t) \cdot P(\text{risky} \mid \lambda) = d - C_t$ gives $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)P_0}{d-C_t}$. At TWAP ($u=1$), $\lambda_0 = \frac{1}{\alpha} \ln \frac{TP_0}{d}$. The multiplicative weight relative to TWAP reduces to $w = \exp(\alpha(\lambda - \lambda_0)) = \frac{(T-t)/T}{(d-C_t)/d} = \tau/\rho = 1/u$. This is a motivating observation, not a guarantee: the Boltzmann assumption does not hold for trained policies. SAUTE [Sootla et al., 2022] identifies the same u as a state-augmentation feature; the closed form here serves only as one default schedule among many monotone candidates.

A.2 Schedule Approximation

A.3 Safety Controllers

APF navigator (Safety Gym). Agent position x, goal g, hazard centres $\{h_i\}$ with radius r, influence distance r_{inf}: $F_{\text{goal}} = (g-x)/\|g-x\|$, $F_{\text{rep}} = k_r \sum_{i:\|x-h_i\|<r+r_{\text{inf}}} (x-h_i)/\|x-h_i\|^3$, $d^\star = \text{norm}(\alpha F_{\text{goal}} + (1-\alpha)F_{\text{rep}})$. Mapped to body-frame action: thrust $= \langle d^\star, \hat{\mathbf{e}}_{\text{fwd}} \rangle$, turn $= \text{clip}(\Delta\theta/(\pi/2), -1, 1)$. We use $\alpha = 0.5$, $r_{\text{inf}} = 0.6$ m, $k_r = 1.0$.

Zero-yaw / WakeAdapt (wind farm). $\pi_{\text{safe}}(s) \equiv 0$ in normalized yaw-action space, minimising the per-turbine surrogate cost. Deployment replaces with a precomputed table $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimising farm-level DEL.

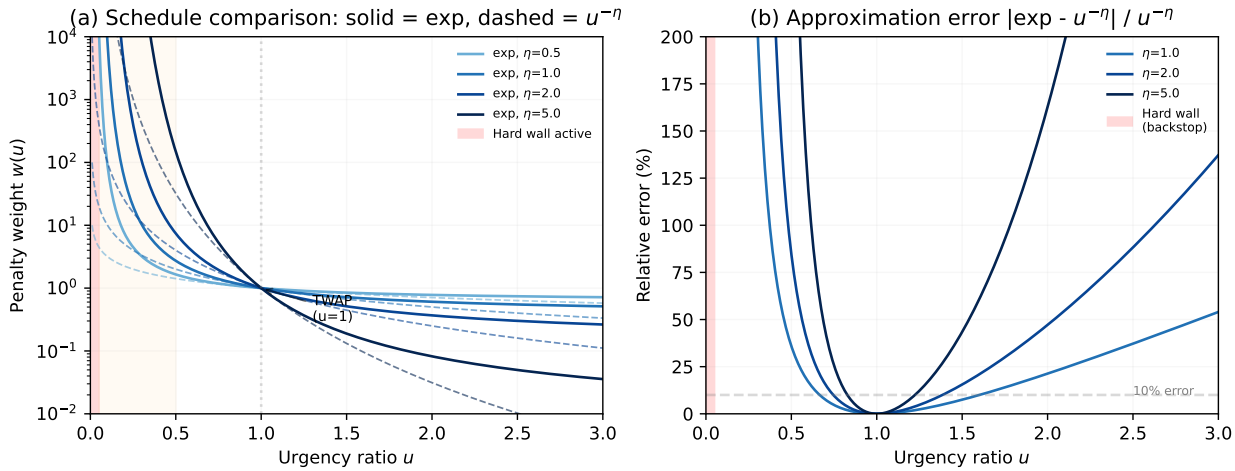

Figure 3: (a) $\exp(\eta(1/u-1))$ (solid) vs. $u^{-\eta}$ (dashed). Red: $u < 0.05$ hard-wall zone. (b) Relative approximation error $< 10\%$ for $u > 0.5$.

A.4 Binary vs Nonlinear Cost (Wind Farm)

Table 10: Binary vs. nonlinear fatigue cost (3-turbine wind farm, 10 ep at 12–16 m/s).

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	[66, 75, 71]	20.3
Binary AC (η=2)	28.1	[8, 8, 10]	86.2
Nonlinear AC (η=5)	**29.2**	[80, 81, 83]	**12.8**

A.5 Learned Cost Critic on Wind Farm

Table 11: Learned Q_c on 3-turbine wind farm (10 ep). Q_c trained on 100 ep with the nonlinear fatigue cost.

Method	Power (MW)	NegYaw	Fatigue	Fatigue ×
Unconstrained	24.6±5.6	[84, 96, 94]	54.6	1.0×
Q_c, η=2, gs=0.05	**28.1±2.7**	[43, 44, 43]	**9.5**	5.7×
Q_c, η=5, gs=0.5	28.8±2.0	[55, 54, 51]	10.4	5.3×
Analytical surrogate (η=5)	29.2	[80, 81, 83]	12.8	4.3×

A.6 Reward–Cost Scatter: from-scratch SAC-Lag (additional ablation)

A.7 Nine Attempted Gradient-Variant Rescues on SG

When the gradient variant is forced to work on SG (low κ), the following all fail:

1. **Pessimistic Q_c ensemble.** 10 heads, $\nabla_a(\bar{Q} + q\sigma_Q)$ with $q \in \{0, 1, 4\}$. $q=0$ ties or wins at every budget.
2. **Online Q_c refit.** Episode-boundary MC refit with L2 anchor; no improvement.
3. **MPC with learned $f_\theta + Q_c$.** Matches unconstrained (55 cost, 17% sat at $d=10$).

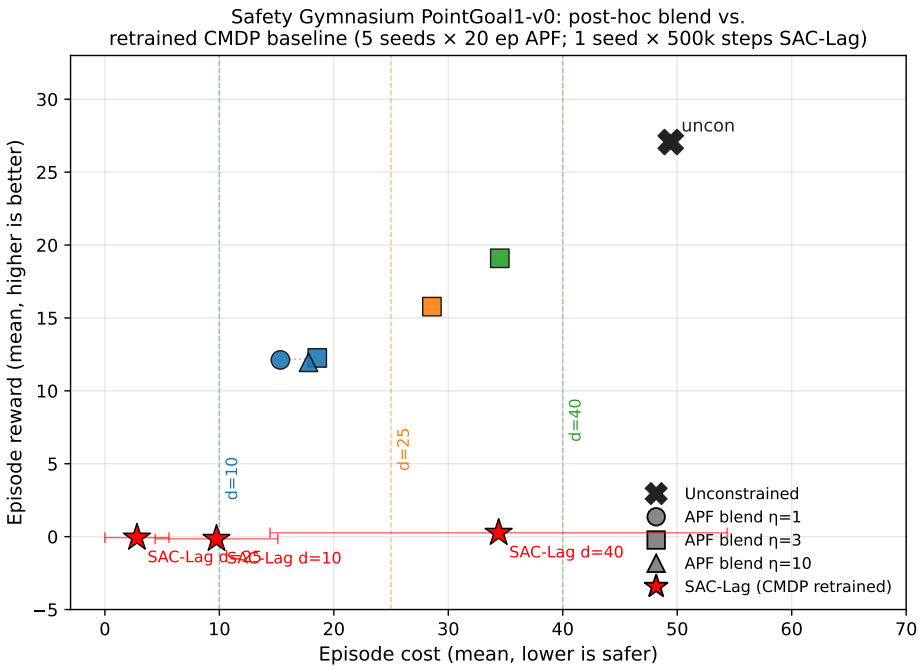

Figure 4: From-scratch SAC-Lag (red stars), 500k steps per budget, vs. APF blend (blue/orange/green) and unconstrained (X). Same qualitative failure mode as the from-pretrained Lagrangian finetune in Figure 2; the choice of warm-start does not rescue CMDP at this compute.

4. **MPC true-env + oracle cost, cost-only.** Reward collapses to -120; 26% sat via "agent stuck in corner."
5. **MPC true-env reward-weighted.** λ (to 10^6) dominates $\beta \cdot r$; same outcome.
6. **MPC true-env feasibility-filtered.** Few feasible candidates; falls back to min-violation; same outcome.
7. **Full MBRL post-hoc.** Ensemble $f_\theta + Q_r + Q_c$ + CEM + online refit. Cost $\approx$ uncon, reward $\approx$ uncon, sat $\approx$ uncon.
8. **CBF 1-step kinematic.** Reward preserved, cost barely drops; 1-step insufficient for momentum.
9. **CBF multi-step (constant-action rollout).** Errors compound; reward degrades, cost rises.

The explicit π_{safe} blend (main body) succeeds because π_{safe} is *not* constrained to lie near π_{perf}'s action distribution. When the performance policy was trained for short-path navigation, its action support does not cover the safe-corridor actions; a local gradient correction cannot reach them. An APF navigator can.

A.8 Coupling-Strength Measurement Protocol

20 episodes per domain under the frozen policy; uniform subsample of 2000 (s, a) pairs; $\|\nabla_a Q_c\|_2$ and $\|\nabla_s Q_c\|_2$ via autodiff; κ = ratio of sample means; nonparametric bootstrap with 1000 resamples for CIs.

A.9 Retraining-Baseline Hyperparameters

SAC-Lagrangian default: $\mu_0 = 0.1$, $\mu_{\text{lr}} = 0.005$, no warmup, no reward scaling. **Warmup variant**: μ held at 0 for the first 50k steps (pure-reward warmup), $\mu_0 = 0.05$ after, $\mu_{\text{lr}} = 0.005$, reward$\times$5. **Aggressive**:

$\mu_0 = 0.5$, $\mu_{\text{lr}} = 0.05$, reward$\times 5$, no warmup. All start from the pretrained unconstrained actor; $\alpha_{\text{ent}} = 0.2$ fixed. **SAUTE**: state-augmentation with ρ_t, reward replaced with $-1/(1-\gamma)$ when $\rho_t < 0$; otherwise standard SAC; trained from the pretrained actor for 500k steps.

A.10 9-Turbine Zero-Shot Transfer Ablation

We zero-shot-transfer the pretrained 3-turbine EBT actor to the 9-turbine 3×3 `square_3x3` layout (the transformer is permutation-equivariant) and re-run a subset of the ablation modes (3 seeds $\times$ 30 ep). Pure π_{safe} retains 97.8% of unconstrained power, mirroring the 3-turbine result; uncon power varies ~ 2 MW seed-to-seed and the wake-steering advantage of π_{perf} is not visible above this variance. The same pattern across both layouts indicates the limitation is in actor training, not farm geometry.

Table 12: Wind-farm 9-turbine zero-shot ablation. Pretrained 3-turbine EBT actor transferred to 3×3 grid. Pattern matches 3-turbine: pure π_{safe} is power-competitive with uncon RL.

Mode	% Uncon power	Blend (MW)	Uncon (MW)	Seeds sat
π_{safe} alone (zero yaw)	97.8±1.2%	76.9	78.7	3/3
const $\sigma = 0.5$	101.8±1.6%	78.0	76.7	0/3
const $\sigma = 0.7$	103.5±2.5%	78.2	75.7	0/3
switch $u < 0.8$	104.2±0.9%	79.2	76.0	3/3
urgency $\eta = 3$	96.9±4.7%	73.9	76.4	3/3

A.11 Path to Wind-Farm Deployment

1. **DEL-calibrated cost**: replace per-turbine surrogate with a control-oriented DEL model [Álvaro-García et al., 2024, Galán-García et al., 2024].
2. **WakeAdapt-equivalent** π_{safe}: precompute $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimising farm DEL.
3. **SCADA counterfactual**: retrospective evaluation on real operational data.
4. **Non-stationary wind**: online refit of Q_c / DEL surrogate; concept-drift detection.
5. **High-fidelity validation**: FAST.Farm calibrated to SCADA.

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